

DATA ANALYTICS PROJECT REPORT

Business Performance 360° Dashboard

Profit & Loss Performance Analysis | Excel Data Preparation • Power BI Dashboard • SQL Query
Library

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September 2026

Tools: Microsoft Excel · Power BI · SQL (PostgreSQL syntax)

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1. EXECUTIVE SUMMARY

This report presents a complete analysis of the Business Performance 360° project, a Profit & Loss (P&L) analytics case study built on a 499-transaction Excel dataset covering the twelve-month period from July 2024 to June 2025, and visualised through an interactive Power BI dashboard. The purpose of the analysis is to evaluate overall business performance across revenue, cost, and profitability, and to translate the underlying transaction-level data into decision-ready insight for business stakeholders.

Across the analysis period, the business generated total revenue of ₹149.16 million, total gross profit of ₹63.37 million (42.5% gross margin), and total net profit of ₹25.10 million (16.83% net margin), after accounting for Cost of Goods Sold, operating expenses, interest, and tax. Revenue was remarkably stable month to month, ranging narrowly between ₹11.70 million and ₹13.34 million — a peak-to-trough swing of only 13.5% — indicating a mature, low-volatility product mix rather than a seasonal or promotion-driven business.

The North region is the strongest performer on every axis measured: it leads in revenue (₹44.31 million), net profit (₹7.64 million), and net margin (17.25%). The East region generates the second-highest revenue (₹39.39 million) but converts it at the weakest net margin of the four regions (16.43%), making it the clearest candidate for a cost review. Product performance is broad-based rather than concentrated: 20 SKUs are active, revenue per product ranges from ₹4.05 million to ₹11.50 million, and storage/accessory products (External SSD 2TB and 1TB) lead the portfolio, while printer and monitor products are disproportionately represented among the 26 individual transactions (5.2% of all transactions) that closed at a net loss.

A key output of this engagement was a rigorous cross-check of the Power BI dashboard against the underlying dataset. That audit surfaced a genuine labelling inconsistency: the dashboard's "Gross Profit" and "EBITDA" KPI cards, and their associated margin cards, are swapped relative to the dataset's own column definitions and relative to the dashboard's own regional Gross Profit chart. This finding, detailed in Sections 6 and 13, is presented as a data-governance recommendation rather than a re-interpretation of the business results, and does not affect the Revenue, Net Profit, or regional figures, which reconcile cleanly throughout.

Overall, the business is financially healthy and performing ahead of its own YTD budget on Revenue, Gross Profit, and Net Profit (each +8.70% versus budget) while holding Operating Expenses 4.76% under budget — a combination that reflects genuine cost discipline rather than under-investment. The recommendations in Section 16 focus on three levers: closing the East region's margin gap, reviewing the printer/monitor product lines that concentrate the loss-making transactions, and resolving the dashboard labelling issue so that KPI cards can be trusted at a glance by senior stakeholders.

2. BUSINESS PROBLEM

Before this dashboard existed, business performance data lived as a flat transaction ledger — 499 individual sales records with no consolidated view of how revenue, cost, and profit were trending, or how performance differed by region, product, or business unit. Senior stakeholders had no single place to answer basic but essential questions: Is the business ahead of or behind budget? Which region is under-performing on margin, not just on revenue? Which products are winning, and which are quietly losing money on individual orders?

The Business Performance 360° Dashboard was built to close that gap. It consolidates the full P&L — Revenue, COGS, Operating Expense, Interest, Tax, Gross Profit, EBITDA, and Net Profit — into a single interactive view, filterable by Region, Year, and Month, so that stakeholders can:

- Monitor overall business performance (revenue, profit, and margin) at a glance via KPI cards.
- Track how monthly revenue and net profit margin are trending across the analysis period.
- Compare regions on revenue, gross profit, and net profit side by side.
- Identify the highest-revenue customers to understand concentration risk.
- Check actual performance against budget across the four core financial metrics.
- Drill down interactively by Region, Year, or Month without needing a fresh data pull.

In short, the dashboard converts a static transaction export into a living management tool — replacing manual spreadsheet reconciliation with a single source of truth that updates as new transactions are added.

3. PROJECT OBJECTIVES

The project was scoped around the following objectives:

- Prepare and validate the raw P&L transaction dataset for analysis (data cleaning and quality checks).
- Analyse overall business performance: total revenue, gross profit, EBITDA, net profit, and margins.
- Monitor how revenue and net profit margin move month to month across the analysis period.
- Compare revenue and profitability across the four operating regions (East, North, South, West).
- Identify the highest- and lowest-performing products by revenue and by profitability.
- Evaluate performance by Business Unit as a secondary organisational lens, distinct from customer Region.
- Assess actual performance against budget across Revenue, Gross Profit, Net Profit, and Operating Expenses.
- Deliver an interactive Power BI dashboard with Region, Year, and Month filtering for self-service exploration.
- Build a companion SQL query library to demonstrate the same analysis reproduced in a relational-database context.
- Independently verify that the dashboard's visuals and KPI cards reconcile against the source data, and flag any discrepancy found.

4. DATASET OVERVIEW

The analysis is built on a single Excel/CSV export named P_L_DATASET, containing 499 transaction-level records across 20 columns. Each row represents one sales transaction with its associated cost and profit breakdown.

4.1 Structure

Attribute	Value
Total records	499 transactions
Total columns	20
Time period covered	15 Jul 2024 – 15 Jun 2025 (rolling 12-month window)
Calendar-year split	251 transactions dated 2024, 248 dated 2025
Regions	4 — East, North, South, West
Business Units	4 — East / North / South / West Region (independent of the Region field — see 4.3)
Distinct products	20 SKUs
Distinct customers	220

4.2 Key Fields

- **TransactionID / Date** — unique transaction key and transaction date.
- **Region / Business Unit** — two independent organisational dimensions (geography vs. internal unit).
- **Product / Customer** — the SKU sold and the purchasing customer.
- **Revenue** — gross sales value, stored as a formatted currency string (₹) rather than a plain number.
- **COGS, Operating Expense, Interest, Tax** — the four cost components that step Revenue down to Net Profit.
- **Gross Profit, EBITDA, Net Profit, Net Profit Margin** — pre-calculated profitability fields.
- **Financial Metric, Actual, Budget** — a budget-comparison reference joined onto each transaction (see 4.3).
- **Month, Year** — calendar attributes used for the time-based visuals.

5. DATA PREPARATION

Before analysis, the dataset was reviewed and validated in Excel to confirm it was fit for use in Power BI and for the SQL query library. The following checks were performed:

- **Missing values:** None. All 20 fields are fully populated across all 499 rows.
- **Duplicate records:** None. TransactionID is unique across all 499 rows, and no full-row duplicates exist.
- **Revenue formatting:** The Revenue column was stored as a formatted text string with a ₹ symbol and thousands separators (e.g. “₹ 2,54,841.00”, using Indian digit grouping). This was cleaned and converted to a numeric field before any aggregation.
- **Date fields:** Dates were verified to be in a consistent DD-MM-YYYY format and parsed successfully with no invalid or out-of-range values.
- **Numeric validation:** COGS, Operating Expense, Interest, Tax, Gross Profit, EBITDA, and Net Profit were confirmed as clean numeric fields with no text contamination.

- **Column/category review:** Region (4), Business Unit (4), Product (20), and Financial Metric (4) were checked for consistent spelling and casing — no inconsistent category labels were found.
- **Negative-value check:** 26 transactions (5.2% of all records) carry a negative Net Profit; these are legitimate loss-making transactions, not data errors, and are analysed in Section 15.

Once validated, the cleaned dataset was loaded into Power BI as the single data source for the dashboard, and the same field structure was mapped to a normalised `pnl_data` table (lower-case, snake_case columns) for the SQL query library in Section 14.

6. KEY PERFORMANCE INDICATORS

The dashboard surfaces six top-level KPI cards. The table below states what each measures, why it matters, and the result observed — using the labels exactly as they appear on the dashboard.

KPI (as labelled on dashboard)	Value	What it measures / why it matters
Total Revenue	₹149.16 M	Gross sales value across all 499 transactions; the top line the business is scaled against.
Gross Profit	₹33.59 M	Labelled as revenue less COGS; the margin retained before operating costs (see audit note below).
EBITDA	₹63.37 M	Labelled as earnings before interest, tax, depreciation & amortisation (see audit note below).
Net Profit	₹25.10 M	The bottom-line profit after COGS, operating expense, interest, and tax — reconciles cleanly to the dataset's Net Profit field.
Gross Profit Margin	16.83	Displayed as Gross Profit ÷ Revenue (see audit note below).
Net Profit Margin	22.52	Displayed as Net Profit ÷ Revenue (see audit note below).

6.1 Dashboard Labelling Audit

As part of verifying every dashboard number against the source dataset, the two profit KPI cards and their corresponding margin cards were cross-checked against (a) the dataset's own Gross Profit and EBITDA columns, and (b) the dashboard's own “Gross Profit by Region” chart, which is built from the dataset's Gross Profit field and sums correctly to ₹63.37 M — not the ₹33.59 M shown on the top KPI card of the same name. The reconciliation is set out below.

Dashboard card	Card value	Reconciles to	Underlying dataset field
“Gross Profit” KPI	₹33.59 M	Dataset EBITDA column total	EBITDA = ₹33,592,310
“EBITDA” KPI	₹63.37 M	Dataset Gross Profit column total (also matches the Gross Profit – by – Region chart)	Gross Profit = ₹63,372,056

Dashboard card	Card value	Reconciles to	Underlying dataset field
"Gross Profit Margin"	16.83	$\text{Net Profit} \div \text{Revenue} = 25.10 \div 149.16$	True Net Profit Margin
"Net Profit Margin"	22.52	$(\text{dataset EBITDA field}) \div \text{Revenue} = 33.59 \div 149.16$	—

7. REVENUE ANALYSIS

Total revenue across the analysis period was ₹149,157,677 (≈₹149.16 M) from 499 transactions, an average of ₹298,913 per transaction. Revenue is highly stable across the twelve months tracked, with no month falling outside a tight ₹11.70 M–₹13.34 M band.

Month	Revenue	Net Profit
Jan	₹13.34 M	₹2.28 M
Feb	₹12.16 M	₹1.98 M
Mar	₹12.30 M	₹2.01 M
Apr	₹12.22 M	₹2.10 M
May	₹13.05 M	₹2.42 M
Jun	₹12.55 M	₹2.18 M
Jul	₹11.70 M	₹1.85 M
Aug	₹12.27 M	₹2.02 M
Sep	₹12.49 M	₹2.03 M
Oct	₹12.49 M	₹2.13 M
Nov	₹11.72 M	₹1.79 M
Dec	₹12.86 M	₹2.30 M

Jan (₹13.34 M) is the single highest-revenue month, followed by May (₹13.05 M) and Dec (₹12.86 M). Jul (₹11.70 M) and Nov (₹11.72 M) are the two softest months. The gap between the strongest and weakest month is only ₹1.64 M — about 13.5% of the weakest month's value — which is too narrow and too irregular in shape to call a clear seasonal pattern; it reads more as ordinary month-to-month variation around a stable baseline than a repeating seasonal cycle.

Because the dataset spans two partial calendar years (Jul–Dec 2024 and Jan–Jun 2025) rather than two full comparable years, a year-over-year growth comparison is not meaningful with this data; the month-over-month view above is the more reliable trend read, and it is reproduced with exact percentage changes in the SQL query library (Section 14, Query 21).

8. PROFIT ANALYSIS

Total net profit across the period was ₹25,097,239 (≈₹25.10 M), a 16.83% net margin on total revenue. Total gross profit was ₹63,372,056, a 42.5% gross margin — meaning cost of goods sold consumes roughly 57.5% of revenue before operating expense, interest, and tax are applied.

Region	Revenue	Net Profit	Net Margin	Gross Margin
North	₹44.31 M	₹7.64 M	17.25%	42.33%
East	₹39.39 M	₹6.47 M	16.43%	42.53%
West	₹33.16 M	₹5.65 M	17.03%	42.47%
South	₹32.30 M	₹5.33 M	16.52%	42.66%

North leads on every profitability measure: highest revenue, highest absolute net profit, and the best net margin (17.25%). East is the standout anomaly — it generates the second-highest revenue of the four regions but converts it at the weakest net margin (16.43%), a 0.82-point gap versus North despite a broadly similar gross margin (42.53% vs. 42.33%). Since gross margins are tight and consistent across all four regions (42.3%–42.7%), the net-margin gap is being driven below the gross-profit line — in operating expense, interest, or tax — rather than in product cost, making East's operating cost base the natural place to start a margin-improvement review.

South has the lowest absolute revenue and profit of the four regions but is not the weakest on margin (16.52%, ahead of East) — its opportunity is one of scale rather than efficiency.

9. PRODUCT PERFORMANCE ANALYSIS

The dataset does not contain a separate Category field — Product is the only item-level dimension available (20 distinct SKUs). This section therefore analyses performance at the individual product level rather than by category, and Section 10 uses Business Unit as the secondary grouping dimension in place of a category hierarchy that the source data does not provide.

9.1 Top-Performing Products by Revenue

Product	Revenue	Net Profit	Net Margin	Transactions
External SSD 2TB	₹11.50 M	₹1.97 M	17.09%	37
External SSD 1TB	₹9.51 M	₹1.62 M	17.00%	33
Desktop Workstation	₹9.31 M	₹1.63 M	17.56%	30
Wireless Mouse	₹8.84 M	₹1.66 M	18.81%	27
Webcam HD	₹8.80 M	₹1.53 M	17.44%	29

Storage and everyday-accessory products lead the portfolio: External SSD 2TB is the single highest-revenue product (₹11.50 M, 37 transactions), with External SSD 1TB close behind. Revenue is genuinely diversified —

the top five products together account for roughly 32% of total revenue, and no single SKU dominates the portfolio, which reduces concentration risk but also means there is no one “blockbuster” product to double down on.

9.2 Lower-Performing Products

Product	Revenue	Net Profit	Net Margin	Transactions
Wireless Keyboard	₹4.05 M	₹0.72 M	17.86%	15
Business Laptop 15	₹4.68 M	₹0.71 M	15.14%	17
4K Monitor 32	₹5.30 M	₹0.72 M	13.57%	20
Laser Printer	₹6.00 M	₹0.95 M	15.83%	20
Ergonomic Mouse	₹6.08 M	₹0.81 M	13.36%	23

Wireless Keyboard has both the lowest revenue and the fewest transactions (15) of any product — it is a genuine long-tail SKU rather than a high-volume, low-price line, and is a candidate for a stocking or bundling review. More significant from a profitability standpoint are Ergonomic Mouse and 4K Monitor 32, which post the two lowest net margins in the portfolio (13.36% and 13.57%) despite mid-pack revenue — both sit roughly 3.5 margin points below the company average of 16.83% and merit a cost-structure review ahead of a revenue-driven fix.

Cross-referencing with the loss-making transactions identified in the data-preparation review (Section 15), Office Printer and Office Monitor 27 each account for 4 of the 26 individual transactions that closed at a net loss, and Laser Printer accounts for a further 3 — printer and monitor lines are disproportionately represented among loss-making orders relative to their share of total transaction volume.

10. BUSINESS UNIT PERFORMANCE ANALYSIS

Business Unit (East / North / South / West Region) is recorded independently of the customer's geographic Region (Section 4.3), so it is analysed here as its own performance lens — most likely reflecting an internal servicing or reporting structure rather than the customer's location.

Business Unit	Revenue	Net Profit
West Region	₹43.57 M	₹7.44 M
North Region	₹38.89 M	₹6.58 M
South Region	₹34.14 M	₹5.80 M
East Region	₹32.56 M	₹5.27 M

Ranked by Business Unit, West Region leads on both revenue and net profit — a markedly different picture from the geographic Region view in Section 8, where West ranks third on revenue behind North and East. This divergence confirms that Business Unit and Region are measuring genuinely different things and should not be used interchangeably in stakeholder reporting; the dashboard and any future reporting should clearly label which of the two dimensions a given chart is sliced by.

11. REGIONAL PERFORMANCE ANALYSIS (GEOGRAPHIC REGION)

This section consolidates the geographic Region view introduced in Section 8, ranking all three profitability lenses side by side.

Rank	By Revenue	By Net Profit	By Net Margin
1st	North (₹44.31 M)	North (₹7.64 M)	North (17.25%)
2nd	East (₹39.39 M)	East (₹6.47 M)	West (17.03%)
3rd	West (₹33.16 M)	West (₹5.65 M)	South (16.52%)
4th	South (₹32.30 M)	South (₹5.33 M)	East (16.43%)

North is the only region to hold 1st place on every measure — it is the benchmark region and a natural source of best practice for the other three. East is the clearest opportunity: it holds onto 2nd place on both revenue and absolute profit, but drops to last place on margin, meaning its profit dollars are real but comparatively inefficient. West and South are close to each other on margin (17.03% vs. 16.52%) despite West generating roughly ₹900K more revenue, and both trail North by a full margin point or more. Because gross margins are within half a point of each other across all four regions (42.3%–42.7%), the margin gaps below Gross Profit — in operating expense, interest, or tax — are the more productive area for a regional cost review than product mix or pricing.

12. MONTHLY SALES TREND ANALYSIS

Month-over-month revenue change across the twelve tracked months (in calendar sequence, Jul 2024 → Jun 2025) is shown below.

Month	Revenue	Change vs. prior month	% Change
Jul 2024	₹11.70 M	—	—
Aug 2024	₹12.27 M	+₹0.57 M	+4.89%
Sep 2024	₹12.49 M	+₹0.21 M	+1.72%
Oct 2024	₹12.49 M	+₹0.001 M	+0.00%
Nov 2024	₹11.72 M	−₹0.76 M	−6.10%
Dec 2024	₹12.86 M	+₹1.13 M	+9.65%
Jan 2025	₹13.34 M	+₹0.49 M	+3.78%
Feb 2025	₹12.16 M	−₹1.19 M	−8.89%
Mar 2025	₹12.30 M	+₹0.15 M	+1.21%
Apr 2025	₹12.22 M	−₹0.08 M	−0.65%
May 2025	₹13.05 M	+₹0.83 M	+6.75%

Month	Revenue	Change vs. prior month	% Change
Jun 2025	₹12.55 M	−₹0.49 M	−3.79%

The strongest single-month gain is Dec 2024 (+9.65%), immediately following the softest month of the period, Nov 2024 (−6.10%) — consistent with ordinary month-to-month variation rather than a sustained trend in either direction. The sharpest decline, Feb 2025 (−8.89%), likewise reverses the prior month's high (Jan 2025, the single strongest month overall). Taken together, the series shows a business oscillating within a stable band rather than trending structurally up or down: no run of three or more consecutive months moves in the same direction, which argues against reading any of these swings as the start of a seasonal cycle without at least one more full year of data to confirm repetition.

13. POWER BI DASHBOARD OVERVIEW

The Business Performance 360° Dashboard is a single-page Power BI report built on the cleaned P&L dataset, with slicers for Region, Year, and Month running across the top. A full-page capture is shown below, followed by a visual-by-visual explanation.



Figure 1. Profit & Loss Dashboard (P&L) — full-page view, Power BI.

13.1 Filters

Filter	Options	Purpose
Region	East, North, South, West	Isolates every visual on the page to a single region for like-for-like comparison.
Year	2024, 2025	Splits the rolling 12-month dataset by calendar year.
Month	Jan – Dec	Focuses every visual on one calendar month across whichever year(s) are selected.

13.2 KPI Cards

Six cards across the top of the report surface Total Revenue, Gross Profit, EBITDA, Net Profit, Gross Profit Margin, and Net Profit Margin. They give a stakeholder the full P&L in one glance, and update live with the Region/Year/Month filters. As documented in Section 6.1, the Gross Profit and EBITDA cards (and their margin counterparts) are cross-wired and should be corrected before wider distribution; Total Revenue and Net Profit reconcile correctly.

13.3 Revenue v/s Trend

A combination bar-and-line chart plotting monthly revenue (bars, labelled) against a secondary trend line. Business question answered: “Is revenue growing, shrinking, or flat, and which months stand out?” Insight: revenue holds in a narrow ₹11.70 M–₹13.34 M band with no runaway growth or decline (Section 7). A stakeholder can use it to spot the softest months (Jul, Nov) for demand-generation focus.

13.4 Net Profit Margin Trend (%)

A lollipop/scatter chart tracking net profit margin percentage by month, ranging from 7.79% (Nov) to 8.96% (Jan) in the chart's own month-level percentage calculation. Business question answered: “Is profitability, not just revenue, holding up month to month?” Insight: margin moves in step with revenue rather than diverging from it, meaning cost ratios are stable rather than eroding profit during high- or low-revenue months. A stakeholder can use this to confirm that a strong revenue month is also a strong profit month, and vice versa.

13.5 P&L Summary

A horizontal bar restating monthly revenue, sorted by month. Business question answered: “What is this month's headline number, read left to right instead of a time-series chart?” This view is most useful as a quick-reference table for a monthly business review meeting, complementing rather than duplicating the trend chart above it.

13.6 Gross Profit / Net Profit / Revenue by Region

Three matching horizontal bar charts, one per metric, each broken out by the four geographic regions. Business question answered: “Which region is winning, and on which metric?” Insight: North leads on all three metrics; East is second on Revenue and Gross Profit but not on Net Profit share in the same proportion, foreshadowing the margin gap quantified in Section 8. A regional manager can use these three charts side by side to see whether their region's profit ranking matches its revenue ranking, or lags behind it as East's does.

13.7 Top 10 Customer by Revenue

A lollipop chart ranking the ten highest-revenue customers, from ₹1.05 M to ₹1.13 M. Business question answered: “Is revenue concentrated in a few key accounts, or spread widely?” Insight: the top 10 customers are tightly clustered (a ₹0.08 M spread across all ten) and together contribute only about 7.3% of total revenue — there is no dominant “whale” account, so account-concentration risk is low, but there is also no anchor customer to build a key-account program around. A sales leader can use this to confirm that customer risk is diversified before setting retention priorities.

13.8 Budget v/s Actual YTD

A grouped bar chart comparing Actual to Budget for four financial metrics: Revenue, Gross Profit, Operating Expenses, and Net Profit. Business question answered: “Are we ahead of or behind our own plan?” Insight: Revenue, Gross Profit, and Net Profit are all running 8.70% ahead of budget, while Operating Expenses are running 4.76% under budget — a consistently favourable position across every metric shown. A finance stakeholder can use this single chart as the headline slide for a YTD budget review.

14. SQL BUSINESS ANALYTICS QUERY LIBRARY

To reproduce the same analysis in a relational-database context, the cleaned dataset was also modelled as a 30-question SQL practice and analysis library (PostgreSQL syntax), covering the identical business questions addressed by the Excel workbook and Power BI dashboard. The Data sheet is imported as a table named `pnl_data`, with columns normalised to lower-case snake_case (`transactionid`, `date`, `region`, `product`, `business_unit`, `customer`, `revenue`, `cogs`, `operating_expense`, `interest`, `tax`, `financial_metric`, `actual`, `budget`, `month`, `year`, `gross_profit`, `ebitda`, `net_profit`, `net_profit_margin`).

14.1 Library Structure

Tier	Questions	Focus
Tier 1 — Core Aggregation	Q1 – Q10	Totals, counts, distinct values, group-by revenue and profit by region/product/business unit/customer, monthly trend, per-transaction averages.
Tier 2 — Profitability & Variance	Q11 – Q18	Margin by region and product, loss-making transactions, high-revenue regions, top customers by profit, revenue contribution %, actual-vs-budget variance.
Tier 3 — Window Functions & Advanced Analysis	Q19 – Q30	Ranking within partitions, month-over-month and year-over-year growth, cumulative revenue, above-average benchmarking, duplicate detection, and a combined regional scorecard.

The full 30-question library with runnable SQL is available as a companion reference document. The queries below are representative examples from each tier, executed against the cleaned dataset, with the resulting values and the business reading of each result.

14.2 Representative Queries — Tier 1

Q1 & Q3. Total revenue and transaction count

```
SELECT SUM(revenue) AS total_revenue FROM pnl_data;  
SELECT COUNT(*) AS total_transactions FROM pnl_data;
```

Result: `total_revenue` = ₹149,157,677 | `total_transactions` = 499.

Q5. Revenue and net profit by region

```
SELECT region, SUM(revenue) AS total_revenue, SUM(net_profit) AS total_net_profit  
FROM pnl_data GROUP BY region ORDER BY total_revenue DESC;
```

region	total_revenue	total_net_profit
North	44,309,367	7,643,824
East	39,388,012	6,471,881
West	33,162,487	5,647,417
South	32,297,811	5,334,117

Q7. Average revenue, COGS, and net profit per transaction

```
SELECT AVG(revenue) AS avg_revenue, AVG(cogs) AS avg_cogs, AVG(net_profit) AS
avg_net_profit
FROM pnl_data;
```

Result: avg_revenue ≈ ₹298,913 | avg_cogs ≈ ₹172,776 | avg_net_profit ≈ ₹50,295 per transaction.

14.3 Representative Queries — Tier 2

Q11. Gross profit margin by region

```
SELECT region, ROUND(100.0 * SUM(gross_profit) / NULLIF(SUM(revenue), 0), 2) AS
gross_profit_margin_pct
FROM pnl_data GROUP BY region ORDER BY gross_profit_margin_pct DESC;
```

region	gross_profit_margin_pct
South	42.66
East	42.53
West	42.47
North	42.33

Gross margin is essentially flat across regions (a 0.33-point spread) — confirming, as in Section 8, that Net Profit differences between regions are driven below the gross-profit line, not by product cost.

Q13. Transactions with a negative net profit

```
SELECT transactionid, region, product, revenue, net_profit
FROM pnl_data WHERE net_profit < 0 ORDER BY net_profit ASC;
```

Result: 26 of 499 transactions (5.2%) return a negative net_profit. Grouping the same 26 rows by product shows Office Printer and Office Monitor 27 each appear 4 times, and Laser Printer 3 times — the printer/monitor lines are over-represented among loss-making orders. By region, West (8) and South (7) account for more than half of the loss-making transactions.

Q17. Actual vs. budget by financial metric

```
SELECT financial_metric, SUM(actual) AS total_actual, SUM(budget) AS total_budget,
       SUM(actual) - SUM(budget) AS variance,
       ROUND(100.0 * (SUM(actual) - SUM(budget)) / NULLIF(SUM(budget), 0), 2) AS
       variance_pct
FROM pnl_data GROUP BY financial_metric ORDER BY variance DESC;
```

financial_metric	variance_pct
Revenue	+8.70%
Gross Profit	+8.70%
Net Profit	+8.70%
Operating Expenses	−4.76%

This matches the dashboard's Budget v/s Actual YTD panel exactly (Section 13.8), and is the query that correctly uses the Actual/Budget reference fields — a month-level version of this question (the original template's Q18, “which months exceeded budget”) is not meaningful for this dataset, since Actual/Budget do not vary by month (Section 4.3).

14.4 Representative Queries — Tier 3

Q20. Top 3 products by revenue within each region

```
WITH product_sales AS (
  SELECT region, product, SUM(revenue) AS total_revenue
  FROM pnl_data GROUP BY region, product),
ranked AS (
  SELECT *, DENSE_RANK() OVER (PARTITION BY region ORDER BY total_revenue DESC) AS
  product_rank
  FROM product_sales)
SELECT * FROM ranked WHERE product_rank <= 3 ORDER BY region, product_rank;
```

Region	#1 product	#2 product	#3 product
East	External SSD 2TB (₹4.25 M)	Laptop Pro 14 (₹3.33 M)	Bluetooth Speaker (₹2.87 M)
North	External SSD 1TB (₹3.77 M)	Laptop Pro 16 (₹3.21 M)	Wireless Mouse (₹2.87 M)
South	Office Printer (₹2.49 M)	Office Monitor 27 (₹2.42 M)	External SSD 1TB (₹2.24 M)
West	External SSD 2TB (₹3.55 M)	Desktop Workstation (₹2.88 M)	Wireless Mouse (₹2.57 M)

Notably, South is the only region where a printer/monitor product is its top revenue driver rather than storage — worth a closer look given the same two product lines' over-representation among loss-making transactions above.

Q21. Month-over-month revenue change

Reproduces the trend in Section 12 using LAG() to compute the prior month's revenue and percentage change; results match Section 12 exactly (e.g. Dec 2024 mom_growth_pct = +9.65%, Feb 2025 = −8.89%).

Q24. Products with net profit margin above the company-wide margin

```
WITH product_margin AS (  
    SELECT product, SUM(net_profit)/NULLIF(SUM(revenue),0) AS margin FROM pnl_data GROUP BY  
    product),  
company_margin AS (SELECT SUM(net_profit)/NULLIF(SUM(revenue),0) AS margin FROM pnl_data)  
SELECT p.product, ROUND(100.0*p.margin,2) AS product_margin_pct  
FROM product_margin p CROSS JOIN company_margin c  
WHERE p.margin > c.margin ORDER BY p.margin DESC;
```

Result: 11 of the 20 products (55%) sit above the company-wide net margin of 16.83%, led by Mechanical Keyboard (18.92%), Wireless Mouse (18.81%), and Bluetooth Speaker (18.78%); the 9 below the line are led downward by Ergonomic Mouse (13.36%) and 4K Monitor 32 (13.57%) — consistent with the product-level findings in Section 9.2.

Q28. Duplicate transaction IDs (data-quality check)

```
SELECT transactionid, COUNT(*) AS occurrence_count  
FROM pnl_data GROUP BY transactionid HAVING COUNT(*) > 1;
```

Result: 0 rows returned — confirms the data-preparation finding in Section 5 that TransactionID is unique across all 499 records.

14.5 Key Insights from the SQL Analysis

- The dataset supports the full 30-question tier structure cleanly, with one documented exception: month-level budget variance (a month-by-month version of Q17) is not meaningful, because Actual/Budget are a fixed YTD reference per financial metric rather than a per-month figure.
- Loss-making transactions (26 of 499) are not randomly distributed — they concentrate in printer and monitor products and in the West and South regions, giving a precise, query-verifiable starting point for a margin investigation rather than a general “review costs” instruction.
- Gross margin is flat within a 0.33-point band across all four regions, which is the SQL-verified confirmation that regional net-margin gaps (Section 8) originate below the gross-profit line.
- Ranking products within each region (Q20) shows South is the only region where a printer/monitor product, not a storage product, is the top revenue driver — a genuine regional difference in product mix that a single company-wide product ranking would have hidden.
- A zero-row result on the duplicate-ID check (Q28) is itself a useful, reportable data-quality finding, not just a housekeeping step — it is the SQL-level evidence backing the data-preparation sign-off in Section 5.

15. KEY BUSINESS INSIGHTS

Insight 1

Finding: Revenue is stable and low-volatility: every one of the twelve tracked months falls within a ₹11.70 M–₹13.34 M band, a peak-to-trough swing of only 13.5%, with no run of three or more consecutive months moving in the same direction.

Business Impact: A stable revenue base is easier to plan and staff against, but it also means there is no single “hot month” to concentrate marketing spend around — growth will need to come from expanding the base rather than riding a seasonal wave.

Recommended Action: Use the current monthly band as the budgeting baseline, and treat any future month that breaks meaningfully outside ₹11.70 M–₹13.34 M as a genuine signal worth investigating, rather than normal noise.

Insight 2

Finding: North is the only region to rank 1st on revenue, net profit, and net margin simultaneously; East ranks 2nd on revenue and net profit but last on net margin (16.43%, 0.82 points behind North) despite a near-identical gross margin.

Business Impact: East's shortfall is a cost-efficiency gap below the gross-profit line, not a pricing or product-mix problem — it represents recoverable margin on revenue the business is already winning.

Recommended Action: Commission a targeted review of East's operating expense, interest, and tax lines against North's, using North's cost ratios as the internal benchmark.

Insight 3

Finding: Gross margin is essentially flat across all four regions (42.33%–42.66%, a 0.33-point spread), confirmed independently by both the Power BI regional chart and the SQL query library (Q11).

Business Impact: Because product-level cost of goods sold is consistent everywhere, any region-specific profitability initiative should target operating cost and overhead, not procurement or supplier pricing.

Recommended Action: Redirect regional margin-improvement efforts toward operating-expense benchmarking rather than COGS renegotiation.

Insight 4

Finding: 26 of 499 transactions (5.2%) closed at a net loss, and these losses concentrate disproportionately in two product lines — Office Printer and Office Monitor 27 (4 transactions each) and Laser Printer (3) — and in the West (8) and South (7) regions.

Business Impact: A small, identifiable slice of the transaction base is quietly eroding overall profitability; because it is concentrated rather than spread evenly, it is addressable with a focused fix rather than a broad one.

Recommended Action: Audit the pricing, discounting, and cost allocation specifically applied to printer and monitor transactions in West and South before extending any blanket cost-cutting measure to the wider catalogue.

Insight 5

Finding: Product revenue is broadly diversified: 20 active SKUs, revenue per product ranging ₹4.05 M–₹11.50 M, with the top five products contributing roughly 32% of total revenue and storage/accessory items (External SSD 2TB and 1TB) leading the portfolio.

Business Impact: No single product dominates the business, which limits concentration risk, but it also means there is no one flagship SKU to anchor a marketing campaign or an inventory-priority decision around.

Recommended Action: Prioritise inventory and marketing investment around the top five products as a group, rather than searching for one product to scale disproportionately.

Insight 6

Finding: The top 10 customers by revenue are tightly clustered (₹1.05 M–₹1.13 M each) and together contribute only about 7.3% of total revenue across 220 distinct customers.

Business Impact: Customer concentration risk is low — losing any single customer would not materially damage the business — but there is also no anchor account large enough to justify a dedicated key-account program on its own.

Recommended Action: Treat customer retention as a broad-based, portfolio-wide programme rather than a small-number account-management exercise.

Insight 7

Finding: The business is running ahead of its own YTD budget on Revenue, Gross Profit, and Net Profit (each +8.70%) while holding Operating Expenses 4.76% under budget — a consistent, favourable pattern across every budgeted metric.

Business Impact: This combination indicates genuine cost discipline paired with stronger-than-planned commercial performance, rather than under-investment suppressing the expense line artificially.

Recommended Action: Use the current budget-vs-actual pattern as supporting evidence when planning next period's targets, and confirm the under-spend on Operating Expenses is not deferring costs that will land in a future period.

Insight 8

Finding: Two of the six dashboard KPI cards (Gross Profit and EBITDA) and their corresponding margin cards do not reconcile to the dataset fields they are named after, while every other visual on the same dashboard — including the regional Gross Profit chart — does reconcile correctly.

Business Impact: A senior stakeholder glancing only at the top KPI row would read the business's cost structure incorrectly (understating true Gross Profit by ₹29.8 M), which could distort a fast investment or pricing decision made from the summary view alone.

Recommended Action: Correct the two KPI card bindings (and their margin cards) before the dashboard is shared beyond this analysis, and add a lightweight KPI-reconciliation check to the dashboard's release checklist going forward.

Insight 9

Finding: Business Unit (an internal four-value field) does not map one-to-one to the geographic Region field — for example, transactions from the East region are spread across all four Business Unit values — and the two dimensions produce materially different regional rankings (West Region leads by Business Unit; North leads by geographic Region).

Business Impact: Any stakeholder report that labels a Business Unit chart simply as “Region” risks stakeholders drawing the wrong regional conclusion from the wrong dimension.

Recommended Action: Clarify the intended business meaning of Business Unit with the data owner, and ensure future dashboard labelling distinguishes it clearly from geographic Region.

16. BUSINESS RECOMMENDATIONS

- **Close the East region margin gap.** Benchmark East's operating expense, interest, and tax ratios against North's, since gross margins are already equivalent across regions (Sections 8, 11, 14.3).
- **Investigate the printer and monitor product lines.** Office Printer, Office Monitor 27, and Laser Printer are over-represented among the 26 loss-making transactions and concentrated in West and South — review pricing and cost allocation on these specific lines before any broader action (Sections 9.2, 14.3, 15).
- **Correct the dashboard KPI card bindings.** Relabel or re-bind the Gross Profit and EBITDA KPI cards (and their margin cards) so the top-row summary matches the detail charts beneath it, and switch the currency format from “\$” to “₹” to match the source data (Section 6.1).
- **Clarify the Business Unit dimension.** Confirm its business meaning with the data owner and label it distinctly from geographic Region in all future reporting, given the two produce different regional rankings (Section 10).
- **Sustain the current budget discipline.** Revenue, Gross Profit, and Net Profit are each 8.70% ahead of budget while Operating Expenses run 4.76% under — use this pattern as the basis for next period's targets rather than resetting budgets to the higher actuals without cause (Sections 6, 14.3).
- **Maintain the diversified product and customer base.** With no single product above 8% of revenue and the top 10 customers contributing only 7.3% combined, continue spreading investment across the top-five product group rather than concentrating on one SKU or account (Sections 9, 13.7, 15).
- **Monitor the monthly revenue band as an early-warning signal.** Treat any month that breaks materially outside the observed ₹11.70 M–₹13.34 M range as a genuine change worth investigating, since the historical pattern shows no sustained trend in either direction (Section 12).

17. CONCLUSION

The Business Performance 360° project demonstrates that the underlying business is financially healthy: ₹149.16 million in revenue converted to a 42.5% gross margin and a 16.83% net margin, with performance running ahead of budget on every profit metric tracked and operating expenses held under budget. North stands out as the benchmark region across every measure, and the product and customer base are both diversified enough to limit concentration risk.

The analysis also identifies two clear, addressable opportunities — East region's below-benchmark margin and the loss-making concentration in printer/monitor transactions — each traced to a specific, query-verifiable cause rather than a general observation. Equally important, the rigorous cross-check between the Excel dataset, the Power BI dashboard, and the SQL query library surfaced a genuine dashboard labelling issue on the two headline profitability KPI cards. Catching and documenting this kind of discrepancy, rather than accepting dashboard numbers at face value, is itself one of the project's key deliverables: it protects the credibility of every other number on the page.

With the KPI card correction applied and the two flagged product/region areas investigated, the Business Performance 360° Dashboard is well-positioned to serve as the business's ongoing, self-service source of truth for revenue, profitability, and budget performance — replacing ad hoc spreadsheet reconciliation with a single, trustworthy, interactive view that supports faster, better-informed decisions.

18. PORTFOLIO PROJECT SUMMARY

Project Name: Business Performance 360° Dashboard

Tools Used: Microsoft Excel, Power BI, SQL (PostgreSQL syntax)

Business Objective: Consolidate a 499-transaction P&L dataset into a single interactive view of revenue, cost, and profitability — filterable by region, year, and month — so stakeholders can monitor performance, compare regions and products, and track actual results against budget without manual spreadsheet reconciliation.

Key Analysis: Cleaned and validated the source transaction data; built KPI, trend, regional, and customer-concentration visuals in Power BI; reproduced the full analysis as a 30-question SQL query library spanning core aggregation, profitability/variance analysis, and window-function-based ranking and benchmarking; and independently cross-checked every dashboard number against the underlying dataset.

Key Findings: ₹149.16 M revenue at a 42.5% gross margin and 16.83% net margin, running 8.70% ahead of budget on every profit metric; North region leads on every measure while East lags on margin despite strong revenue; 26 transactions (5.2%) are loss-making and concentrate in printer/monitor products; and a genuine labelling swap was identified and documented on two of the dashboard's six headline KPI cards.

Business Value: Gives stakeholders a single, interactive, filterable view of business performance in place of static spreadsheet reporting, while the documented data-quality and dashboard-labelling audit demonstrates the kind of verification discipline that keeps a self-service dashboard trustworthy at scale.

19. SKILLS DEMONSTRATED

- Microsoft Excel — data cleaning, currency-string parsing, and transaction-level validation
- Data Cleaning & Data Preparation
- Exploratory Data Analysis
- Power BI — dashboard development, KPI cards, slicers, and interactive filtering
- Data Visualisation & Dashboard Development

- SQL — aggregation, CTEs, window functions (RANK, DENSE_RANK, ROW_NUMBER, LAG), and benchmarking queries
- KPI Development and Business Analysis
- Interactive Reporting
- Data Storytelling and Business Insight Generation
- Data-Driven Decision Making
- Cross-source data verification and dashboard quality auditing